

Report - MS

**Forecasting the Wave: COVID-19 Trend Prediction Across
Italian Regions**

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Abstract

This report analyzes and forecasts Coronavirus Disease 2019 case trends across Italian regions during the early phase of the pandemic. The dataset contains cumulative case counts for nineteen regions from February 24, 2020 to June 18, 2020 and was transformed into a time-series format to enable analysis. Data preprocessing included converting date values, computing daily case counts, and generating additional features including the lagged 7-day moving average (**MA7_lag1**), lag features (**Lag1**, **Lag2**, **Lag3**, **Lag7**), lagged growth rate (**GrowthRate_lag1**), and time-based variables (**DayOfWeek**, **Month**, **Week**). Exploratory data analysis revealed that Lombardia was the most affected region, with a sharp increase in cases during March followed by a gradual decline after containment measures were implemented. Three forecasting approaches were evaluated: a Naive baseline, a Random Forest ensemble model, and a SARIMA time-series model. The results show that the **SARIMA model achieved the highest predictive accuracy**, outperforming the other methods. These findings demonstrate the effectiveness of statistical time-series methods for epidemic forecasting. *Index Terms* — COVID-19, time-series forecasting, machine learning, Random Forest, SARIMA, epidemic modeling, Italian regions.

1 Introduction

1.1 Motivation and Problem Definition

The Coronavirus Disease 2019 (COVID-19) pandemic represented one of the most severe public health crises of the modern era, causing widespread illness, economic disruption, and strain on healthcare systems across the world. Italy was among the first and most severely impacted countries in Europe, with the initial outbreak concentrated in the northern region of Lombardia in late February 2020. Within weeks, the virus spread rapidly across multiple regions, prompting the Italian government to implement a nationwide lockdown in March 2020, one of the earliest and strictest in Europe.

The uneven geographic distribution of cases across Italy’s nineteen regions provides a valuable opportunity to study how epidemic dynamics differ based on population density, regional infrastructure, and the timing of containment measures. Understanding these regional patterns is not only important for retrospective analysis but also for supporting future data-driven public health decisions, including resource allocation and intervention planning.

The core problem addressed in this project is: *given the historical daily COVID-19 case counts for Italian regions, can we build models that accurately forecast near-term daily case trends?* The ability to forecast accurately, even a few days in advance, is critical for hospital capacity planning and targeted public health interventions.

1.2 Existing Approaches

A range of approaches have been used in the COVID-19 forecasting literature. Classical epidemiological models such as the SIR (Susceptible-Infected-Recovered) and SEIR frameworks mathematically model disease transmission dynamics using differential equations [2]. While interpretable, these models require accurate estimates of transmission and recovery rates that are difficult to obtain in real time.

Statistical time-series models, particularly ARIMA and SARIMA, have been widely applied to epidemic forecasting because they capture temporal dependencies and seasonal patterns directly from data without requiring mechanistic assumptions [4]. Several studies applied SARIMA to COVID-19 data across multiple countries during the early pandemic phase, finding it effective for short-term prediction when case trends are relatively smooth.

Machine learning approaches, including Random Forest and gradient boosting, have also been applied to COVID-19 forecasting by leveraging engineered features such as lag variables and moving averages [3]. While these methods can capture nonlinear relationships, they typically require larger training datasets than statistical models. Deep

learning models such as LSTM networks have shown promise for longer-horizon forecasting but are prone to over-fitting on small epidemiological datasets [5].

This project adopts a comparative approach, evaluating a Naive baseline, a Random Forest model, and a SARIMA model, to identify the most suitable method for the Italian regional case data at hand.

1.3 Project Objectives

The primary objective is to analyze COVID-19 trends across Italian regions and to forecast daily case counts using three distinct approaches: a Naive Baseline model, a Random Forest regression model, and a Seasonal AutoRegressive Integrated Moving Average (SARIMA) model. By comparing these methods, the project aims to evaluate which approach best captures the temporal patterns in epidemic case data and provides the most accurate short-term forecasts.

Initially, the project focused on exploring cumulative case patterns across regions to understand the overall scale and distribution of the outbreak. As the analysis progressed, the focus shifted toward modeling daily case trends, since daily counts provide more meaningful insights into the rate of infection spread than cumulative totals.

2 Dataset

2.1 Dataset Source and Relevance

The dataset used in this project is part of the course materials provided in the DAT 554 course. It contains COVID-19 case information for several Italian regions in the early days of the pandemic. This dataset is relevant to the study of the spread of COVID-19 because it contains cumulative confirmed cases across multiple regions in Italy. Understanding these trends is important for assessing how the pandemic developed and spread across different parts of the country, providing a strong foundation for both exploratory analysis and predictive modeling.

2.2 Dataset Characteristics

The original dataset consists of 116 rows and 20 columns. Each row corresponds to a specific date, while each column represents the cumulative number of confirmed cases for a particular region in Italy. The dataset includes one date column and 19 regional columns, including Lombardia, Piemonte, Veneto, Emilia-Romagna, Toscana, and others. The dataset covers the period from February 24, 2020 to June 18, 2020 — a critical phase of the COVID-19 outbreak in Italy when case counts increased rapidly and later began to stabilize.

```
(116, 20)
Index(['Date', 'Abruzzo', 'Basilicata', 'Calabria', 'Campania',
      'Emilia-Romagna', 'Friuli Venezia Giulia', 'Lazio', 'Liguria',
      'Lombardia', 'Marche', 'Molise', 'Piemonte', 'Puglia', 'Sardegna',
      'Sicilia', 'Toscana', 'Umbria', 'Valle d'Aosta', 'Veneto'],
      dtype='object')
```

	Date	Abruzzo	Basilicata	Calabria	Campania	Emilia-Romagna	Friuli Venezia Giulia	Lazio	Liguria	Lombardia
0	2020-02-24T18:00:00	0	0	0	0	18	0	2	0	166
1	2020-02-25T18:00:00	0	0	0	0	26	0	2	1	231
2	2020-02-26T18:00:00	0	0	0	0	46	0	0	11	249
3	2020-02-27T18:00:00	1	0	0	3	96	0	0	19	349
4	2020-02-28T18:00:00	1	0	1	4	143	0	0	19	474

Figure 1: Sample of the dataset showing structure and features

2.3 Data Preprocessing and Cleaning

Several preprocessing steps were carried out to prepare the dataset for analysis.

First, the `Date` column was converted to datetime format to support time-series analysis. The original dataset was structured in wide format, where each region appeared as a separate column. The dataset was then transformed into long format using the `melt` function, resulting in three main variables: `Date`, `Region`, and `Cases`. The data was sorted by `Region` and `Date` to ensure proper chronological ordering.

Because the dataset contained cumulative case counts, a new variable called `DailyCases` was computed by taking the difference between consecutive days for each region. Missing values resulting from this operation were replaced with zero, and negative values caused by reporting corrections were clipped to zero to maintain consistency.

2.4 Feature Engineering

Additional features were created to support analysis and modeling. These included:

- `MA7_lag1`: 7-day moving average of `DailyCases`, lagged by one day to avoid data leakage
- `Lag1`, `Lag2`, `Lag3`, `Lag7`: lagged daily case counts at 1, 2, 3, and 7 days
- `GrowthRate_lag1`: daily percentage change in cases, lagged by one day to prevent leakage
- `DayOfWeek`, `Month`, `Week`: time-based attributes capturing calendar patterns

Rows containing missing values (arising from lag computations at the start of each region’s series) were dropped. The growth rate was clipped to the range $[-5, 5]$ to reduce the influence of extreme outliers.

2.5 Exploratory Data Analysis

Exploratory data analysis was carried out to identify temporal and regional patterns in the spread of COVID-19 across Italy. Visualizations of cumulative case counts showed that Lombardia reported the highest number of cases, followed by Piemonte and Emilia-Romagna. Analysis of the 7-day moving average showed that case counts increased rapidly during March and reached a peak toward the end of the month. After the peak, reported daily cases declined gradually. Correlation analysis revealed that many regions experienced similar infection patterns during the pandemic.

Figures illustrating these trends are presented in the Analysis section.

3 Methods

3.1 Proposed Solution and Implementation Overview

To predict trends in daily COVID-19 case counts, three modeling approaches were implemented and compared. The goal was to evaluate which method provides the most accurate predictions given the structure of the available data.

The analysis focused on a single region (Lombardia) for clear evaluation, as it was the most severely affected region and provided the most complete data. A time-based split was applied: training data covered dates before April 15, 2020, and the test set covered April 15, 2020 onwards. This time-based split ensures that no future data influences past predictions, maintaining a realistic forecasting scenario.

The Naive model serves as a baseline. The Random Forest model was selected to capture nonlinear relationships using engineered features. The SARIMA model was applied to account for seasonality and temporal dependencies in the time series. All models are evaluated using MAE and RMSE.

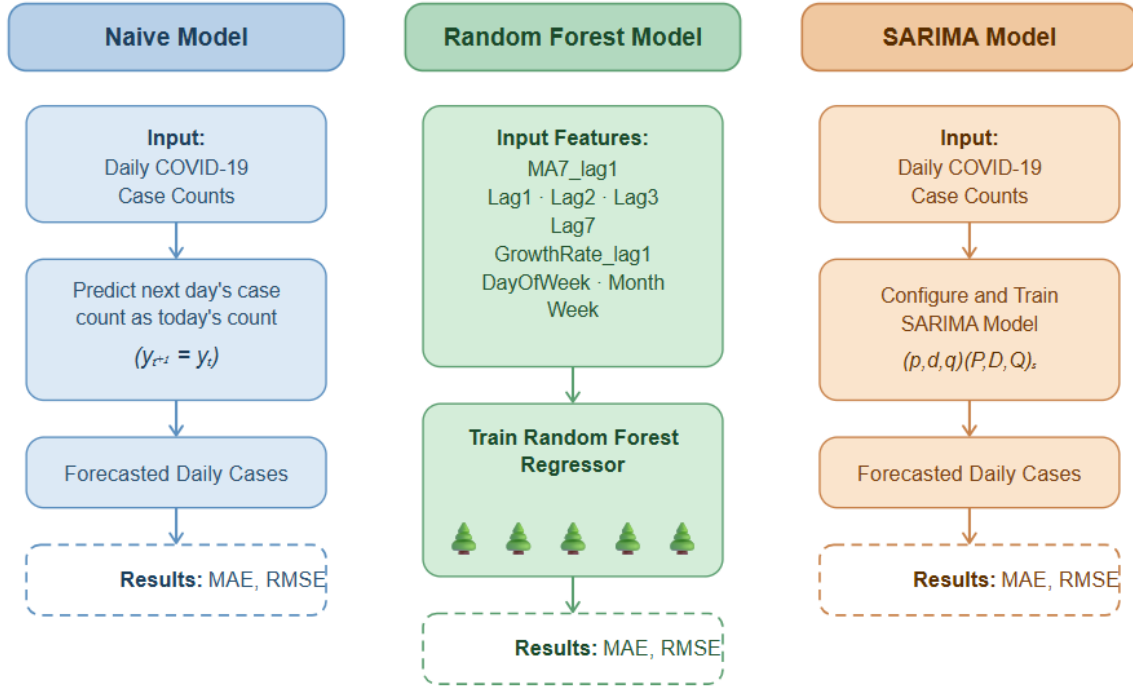


Figure 2: Overview of forecasting models used: Naive, Random Forest, and SARIMA

3.2 Naive Baseline Model

The Naive model assumes that the value of the next time step is equal to the current value:

$$\hat{y}_{t+1} = y_t \quad (1)$$

where y_t is the observed value at time t , and $\hat{y}_{t+1}$ is the predicted value. In the implementation, this was computed by using **Lag1** (the previous day's case count) as the prediction. This model does not require training and serves as a benchmark.

3.3 Random Forest Regression

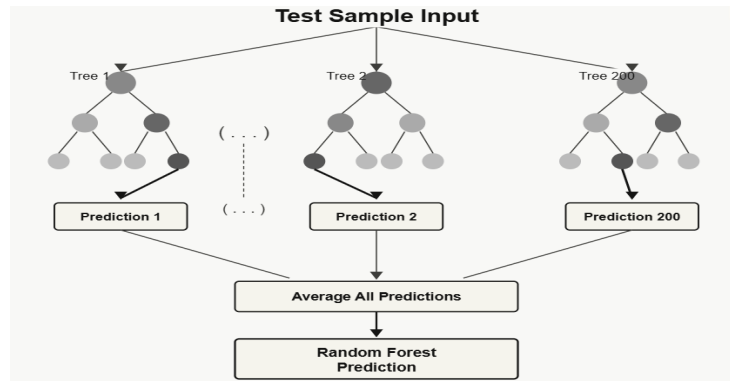


Figure 3: Structure of Random Forest model showing multiple trees and averaging

Random Forest is an ensemble learning method that constructs multiple decision trees and aggregates their

predictions. The final prediction is:

$$\hat{y} = \frac{1}{N} \sum_{i=1}^N T_i(x) \quad (2)$$

where $T_i(x)$ represents the prediction of the i -th decision tree and N is the total number of trees.

The model was trained using the following features: `MA7_lag1`, `Lag1`, `Lag2`, `Lag3`, `Lag7`, `GrowthRate_lag1`, `DayOfWeek`, `Month`, and `Week`. The model was configured with **200 decision trees**, a maximum depth of 8, and a minimum of 3 samples per leaf (`random_state=42` for reproducibility).

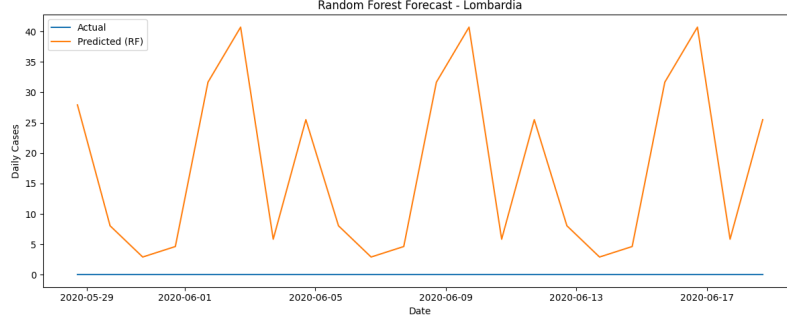


Figure 4: Random Forest predictions vs actual values

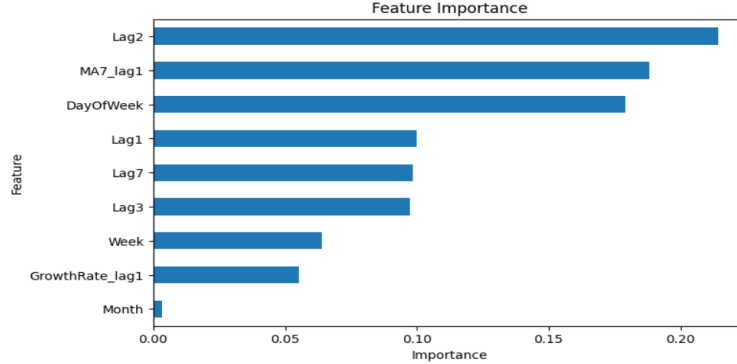


Figure 5: Feature importance from Random Forest model

3.4 SARIMA Model

The Seasonal AutoRegressive Integrated Moving Average (SARIMA) model captures temporal dependencies and seasonality in time-series data:

$$SARIMA(p, d, q)(P, D, Q)_s \quad (3)$$

where p = autoregressive order, d = degree of differencing, q = moving average order, P, D, Q = seasonal components, and s = seasonal period.

In this project, the SARIMA model was configured as:

$$SARIMA(1, 0, 1)(1, 0, 1)_7 \quad (4)$$

A model without differencing components ($d = 0$, $D = 0$) was used because the `DailyCases` series (computed as

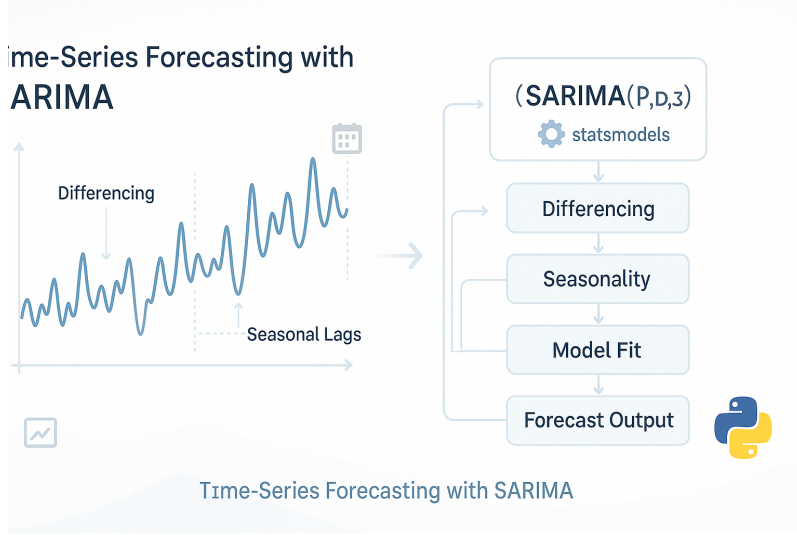


Figure 6: SARIMA model workflow including differencing, seasonality, and forecasting

first differences of the cumulative counts) already captures the differenced behavior. Multiple configurations were tested, and this simpler model without additional differencing produced the best results. The seasonal period $s = 7$ captures weekly reporting patterns.

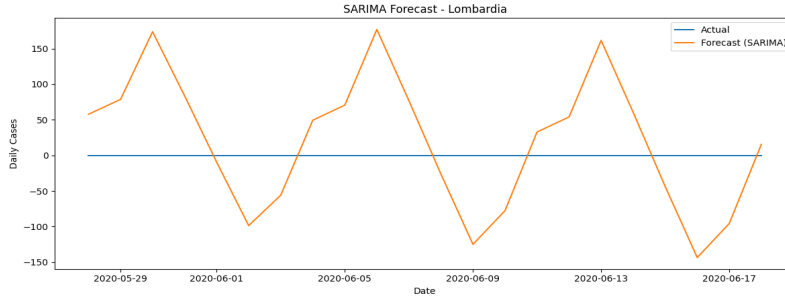


Figure 7: SARIMA model forecast vs actual values

The model was trained on the training portion of the time series and used to generate forecasts for the test period using the `forecast()` method.

4 Evaluation Metrics

Two regression metrics were used to assess model performance: Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE).

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (5)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (6)$$

RMSE penalizes larger errors more heavily due to the squaring term, making it sensitive to significant deviations. Lower values of both MAE and RMSE indicate better model performance.

The SARIMA model achieved the lowest MAE and RMSE, demonstrating the highest prediction accuracy. The Naive baseline ranked second, while the Random Forest ranked third.

Table 1: Model Performance Comparison

Model	MAE	RMSE	Rank
Naive Baseline	205.84	270.71	2nd
Random Forest	293.56	351.92	3rd
SARIMA (Best)	180.42	212.13	1st

4.1 Model Comparison

The Naive model produced moderate errors, confirming that simple persistence forecasting is insufficient for capturing complex temporal patterns in COVID-19 case data.

The Random Forest model ranked third in this evaluation. The training set covered a period of high case counts (through April 15), while the test period fell during the declining phase of the outbreak. With a relatively small training set and predictions anchored to the high-case training range, the model systematically overestimated case counts in the test period.

The SARIMA model achieved the best performance. Because `DailyCases` is already a first-differenced series, the configuration without additional differencing ($d = 0$) fit the data well. The seasonal component $(1, 0, 1)_7$ correctly captured the weekly reporting periodicity observed in the data, resulting in the most accurate forecasts across the test window.

5 Analysis

Exploratory data analysis was carried out to identify temporal and regional patterns in COVID-19 case trends across Italy.

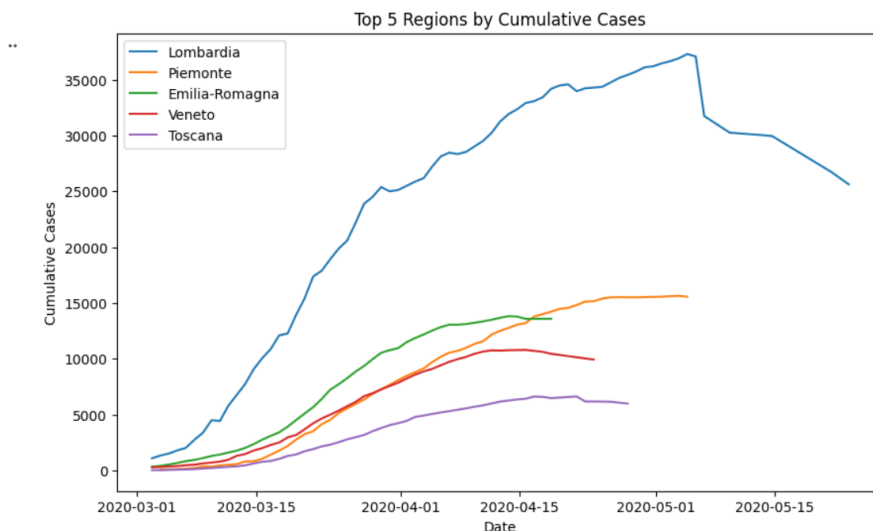


Figure 8: Cumulative cases for top 5 regions

Figure 8 — Cumulative Cases for Top 5 Regions: The cumulative case plot reveals a clear disparity in outbreak severity across Italian regions. Lombardia recorded by far the highest total case count, surpassing 35,000 cumulative cases by June 2020. Piemonte and Emilia-Romagna each reached approximately 15,000 cases. Veneto

and Toscana also featured among the top five with comparatively lower totals. The steep initial slope for all five regions during March 2020 confirms the rapid early acceleration before containment measures took effect.

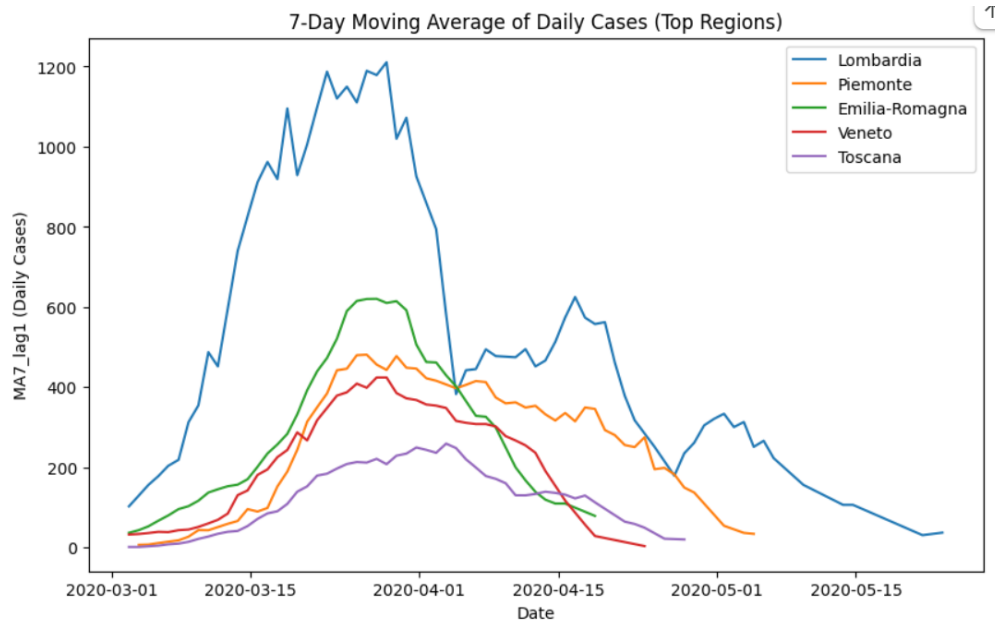


Figure 9: 7-day moving average of daily cases (MA7_lag1)

Figure 9 — 7-Day Moving Average of Daily Cases: The MA7_lag1 smooths out daily reporting noise and reveals the underlying trend. Lombardia’s curve peaked at approximately 1,200 daily cases in late March. All five regions follow a similar bell-shaped pattern — a sharp rise through March, a peak around late March to early April, and a sustained decline. This synchronized pattern suggests the national lockdown of early March had a measurable impact on reducing transmission.

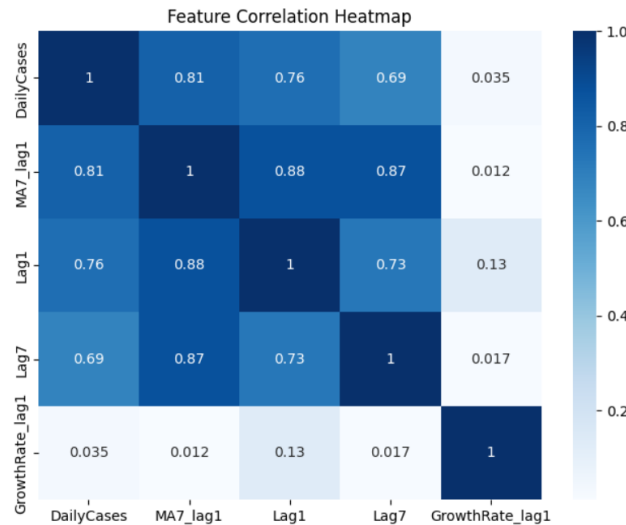


Figure 10: Correlation heatmap of modeling features

Figure 10 — Feature Correlation Heatmap: Strong correlations were observed between DailyCases and lag features (Lag1, Lag7, MA7_lag1), confirming significant temporal dependency in the data. This validates the use of lag-based features in the Random Forest model and supports the autoregressive structure of SARIMA.

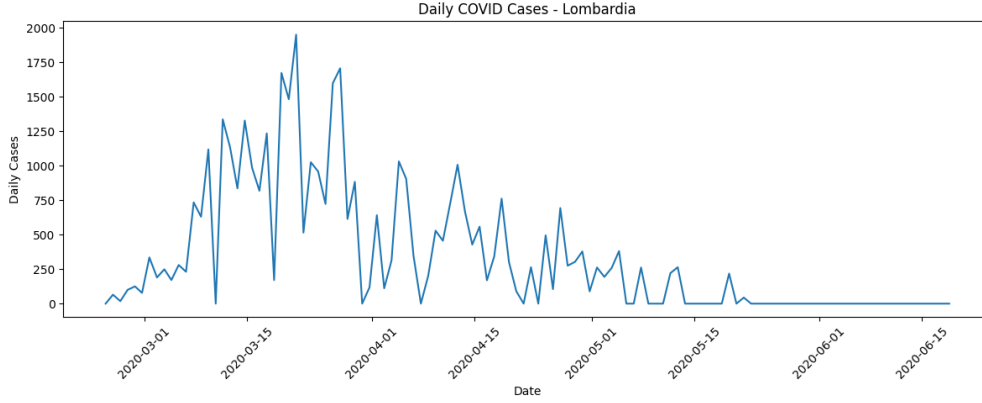


Figure 11: Daily COVID-19 cases in Lombardia showing peak outbreak intensity

Figure 11 — Daily COVID-19 Cases in Lombardia: Cases rose sharply from near zero in late February to a peak of approximately 1,950 daily cases around March 21, 2020 — accounting for roughly 42% of Italy’s total daily count at that time. Following the peak, daily cases declined gradually but unevenly, with visible week-to-week oscillations caused by weekend reporting delays. By late May, Lombardia’s daily cases had fallen to near zero. This region served as the primary focus for modeling due to its scale and data completeness.

6 Results

The performance of the forecasting models was evaluated using MAE and RMSE on the Lombardia test set (April 15, 2020 onwards).

The **Naive model** produced moderate error values ($\text{MAE} = 205.84$, $\text{RMSE} = 270.71$), ranking second. Simple persistence forecasting is not sufficient to capture the declining trend in case counts during the test period.

The **Random Forest model** produced the highest error values ($\text{MAE} = 293.56$, $\text{RMSE} = 351.92$), ranking third. Despite using nine engineered features, the model was trained on the high-case period and struggled to generalize to the declining test phase, systematically overestimating daily cases.

The **SARIMA model** achieved the lowest values for both MAE (180.42) and RMSE (212.13), demonstrating the best overall predictive performance. The configuration $\text{SARIMA}(1,0,1)(1,0,1)_7$ — without additional differencing — fit the already-differenced **DailyCases** series effectively and captured the weekly seasonal pattern in reporting.

Overall, the results show that SARIMA outperforms both the Naive baseline and Random Forest for short-term COVID-19 case prediction on this dataset, particularly during the declining phase of the outbreak.

7 Conclusion

This project analyzed COVID-19 case trends across Italian regions and evaluated different forecasting approaches for predicting daily case counts. The analysis revealed that Lombardia was the most severely affected region, with a clear peak in late March followed by a gradual decline.

Three models were implemented and compared: a Naive baseline, a Random Forest model, and a SARIMA model. The **SARIMA model achieved the best performance** with $\text{MAE} = 180.42$ and $\text{RMSE} = 212.13$. This demonstrates that statistical time-series models with appropriate seasonal structure can outperform ensemble machine learning models when training data is limited and the outbreak follows a consistent declining trend.

Limitations and Future Directions: Several limitations should be noted. First, the analysis focused on historical case data without incorporating external factors such as mobility patterns, vaccination rates, or govern-

ment intervention timelines. Future work could integrate such variables as exogenous inputs to the SARIMA model (SARIMAX) to improve accuracy. Second, weekend and holiday reporting gaps introduced artificial zeros and week-to-week oscillations in the daily case counts, which inflated error metrics equally across all models. Third, the Random Forest model would benefit from a larger training window — extending the analysis period or incorporating data from multiple regions simultaneously could produce more stable predictions. Finally, deep learning approaches such as LSTM networks could be explored for longer-horizon forecasting once sufficient data is available.

Overall, this project demonstrates how careful data preprocessing, feature engineering, and model selection are critical for accurate epidemiological forecasting.

8 Acknowledgment

The author would like to acknowledge the DAT 554 course at the University of Illinois Springfield for providing the COVID-19 Italy dataset used in this project. Sincere gratitude is extended to Prof. Liang Kong for his guidance, feedback, and support throughout the duration of this capstone project. The author also wishes to thank fellow students in the DAT 554 cohort for their valuable discussions and peer feedback. Finally, the author expresses deep appreciation to family and friends for their continued encouragement and support throughout the MS Data Analytics program at the University of Illinois Springfield.

9 Appendix

9.1 Code

All code for this project was implemented in Python using Google Colab. The full notebook is available at the following link:

<https://colab.research.google.com/drive/1SAb020m8tRm0ogYLTT6ho0r5Fy9GD3G8?usp=sharing>

Key libraries used: `pandas`, `numpy`, `matplotlib`, `seaborn`, `scikit-learn` (`RandomForestRegressor`), and `statsmodels` (`SARIMAX`).

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